

STABILITY OF ALGORITHMS

condition number of a problem $\rightarrow$ good/bad problem

stability of an algorithm $\rightarrow$ good/bad algorithm

Rmk: cannot even ask the computer to compute $y = f(x)$

Floating point numbers: double (64 bits)

$$\pm 1. \underbrace{0110 \dots 0110}_{52 \text{ bits}} \cdot 2^{\pm \underbrace{1001 \dots 01}_{10 \text{ bits}}} \quad + \text{special "numbers" like } 0, \text{inf, NaN}$$

$\Rightarrow$ there are "only" 2^{64} numbers that you can represent
 $\in [10^{-308}, 10^{308}]$

but they are not equispaced:

- between 1 and 2 you have 2^{52} numbers
- between 2 and 4 you have 2^{52} numbers
-

$$\left(\frac{1}{10}\right)_{10} = 0.0 \overline{0011}$$

Theorem: For each $x \in \pm [10^{-308}, 10^{308}]$ there's a number $\tilde{x}$ which is representable EXACTLY st.

$$\frac{|\tilde{x} - x|}{|x|} \leq u, \quad u = 2^{-52} \approx 2 \cdot 10^{-16}$$

$\hookrightarrow$ machine precision

For a computational problem, the best we could ask for is (not $y = f(x)$)

$$\tilde{y} = f(\tilde{x}), \quad \text{where } \frac{|\tilde{x} - x|}{|x|} \leq u$$

$$\frac{|\tilde{y} - y|}{|y|} \leq \kappa_{\text{rel}}(f, x) \cdot \underbrace{\frac{|\tilde{x} - x|}{|x|}}_{\text{intrinsic error}} \leq \kappa_{\text{rel}}(f, x) \cdot u$$

Def: An algorithm is stable (on some input x) if the computed output $\tilde{y}$ satisfies $\frac{|\tilde{y} - y|}{|y|} \leq \kappa(f, x) \cdot \underbrace{O(u)}_{\hookrightarrow u \cdot \text{poly}(\text{dimension of the problem}) + O(u^2)}$

Example: $a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad y = a^T b = a_1 b_1 + a_2 b_2 + a_3 b_3$

Let's assume that $a_1, \dots, b_3$ are numbers that can be represented exactly with 64 bits.

$$a_1 \odot b_1 = a_1 b_1 (1 + \delta_1) \quad |\delta_1| \leq u$$

$$a_2 \odot b_2 = a_2 b_2 (1 + \delta_2) \quad |\delta_2| \leq u$$

$$a_3 \odot b_3 = a_3 b_3 (1 + \delta_3) \quad |\delta_3| \leq u$$

$$\tilde{y} = (a_1 \odot b_1 \oplus a_2 \odot b_2) \oplus a_3 \odot b_3$$

$$= (a_1 b_1 (1 + \delta_1) \oplus a_2 b_2 (1 + \delta_2)) \oplus a_3 b_3 (1 + \delta_3)$$

$$= (a_1 b_1 (1 + \delta_1) + a_2 b_2 (1 + \delta_2)) (1 + \delta_4) \oplus a_3 b_3 (1 + \delta_3) \quad |\delta_4| \leq u$$

$$= \left[\underbrace{(a_1 b_1 (1 + \delta_1) + a_2 b_2 (1 + \delta_2))}_{(1 + \delta_4)} + \underbrace{a_3 b_3 (1 + \delta_3)}_{(1 + \delta_5)} \right] (1 + \delta_5) \quad |\delta_5| \leq u$$

$$= a_1 b_1 (1 + \delta_1 + \delta_4 + \delta_5) + a_2 b_2 (1 + \delta_2 + \delta_4 + \delta_5) + a_3 b_3 (1 + \delta_3 + \delta_5) + O(u^2)$$

$$|\tilde{y} - y| = |\tilde{y} - (a_1 b_1 + a_2 b_2 + a_3 b_3)|$$

$$= |a_1 b_1 (\delta_1 + \delta_4 + \delta_5) + a_2 b_2 (\delta_2 + \delta_4 + \delta_5) + a_3 b_3 (\delta_3 + \delta_5)| + O(u^2)$$

$$\leq |a_1 b_1| \cdot (|\delta_1| + |\delta_4| + |\delta_5|) + |a_2 b_2| (|\delta_2| + |\delta_4| + |\delta_5|) +$$

$$+ |a_3 b_3| \cdot (|\delta_3| + |\delta_5|) + O(u^2)$$

$$\leq 3u |a_1| \cdot |b_1| + 3u |a_2| |b_2| + 2u |a_3| \cdot |b_3|$$

$$\leq 3u \underbrace{|a|^\top |b|}$$

$$|a| = \begin{bmatrix} |a_{11}| \\ |a_{21}| \\ |a_{31}| \end{bmatrix} \quad |b| = \begin{bmatrix} |b_1| \\ |b_2| \\ |b_3| \end{bmatrix}$$

↳ this could be much larger than $y = a^\top b$:
for example:

$$y = [1 \ 1 \ -1] \begin{bmatrix} 10^6 \\ 10^6 \\ 10^6 \end{bmatrix} = 1$$

$$[1 \ 1 \ 1] \begin{bmatrix} 10^6 \\ 10^6 \\ 10^6 \end{bmatrix} = 1 + 2 \cdot 10^6$$

Def: Backward stability: sometimes we can see the inexact computed $\tilde{y}$ as the exact solution $\tilde{y} = f(\hat{x})$ for an input $\hat{x}$ which is close to the input x you were interested in.

$$\tilde{y} = f(\hat{x}) \text{ with small } |x - \hat{x}|$$

$$\tilde{y} = \underbrace{a_1 \hat{b}_1 (1 + \delta_1 + \delta_4 + \delta_5)}_{\hat{a}} + \underbrace{a_2 \hat{b}_2 (1 + \delta_2 + \delta_4 + \delta_5)}_{\hat{b}} + \underbrace{a_3 \hat{b}_3 (1 + \delta_3 + \delta_5)}_{\hat{b}_3} + O(u^2)$$

$$\hat{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\hat{b} = \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = \begin{bmatrix} b_1 (1 + \delta_1 + \delta_4 + \delta_5) \\ b_2 (1 + \delta_2 + \delta_4 + \delta_5) \\ b_3 (1 + \delta_3 + \delta_5) \end{bmatrix}$$

$$\|\hat{b} - b\| = \left\| \begin{bmatrix} (\delta_1 + \delta_4 + \delta_5)b_1 \\ (\delta_2 + \delta_4 + \delta_5)b_2 \\ (\delta_3 + \delta_5)b_3 \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} 3u|b_1| \\ 3u|b_2| \\ 2u|b_3| \end{bmatrix} \right\| \leq 3u \|b\|$$

⇒ we showed that the value $\tilde{y}$ computed by the computer is the EXACT inner product of two vectors $\hat{a}, \hat{b}$ s.t.

$$\frac{\|\hat{a} - a\|}{\|a\|}, \frac{\|\hat{b} - b\|}{\|b\|} \leq O(u)$$

Example of non-backward stable algorithm

$$(a, b \in \mathbb{R}^3) \mapsto ab^T$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} [b_1 \ b_2 \ b_3] = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \odot [1 \ 2 \ 3] = \begin{bmatrix} 1.01 & 2.05 & 2.99 \\ 2 & 4.01 & 6.02 \\ 2.97 & 6 & 9.01 \end{bmatrix} = \tilde{y}$$

Q: are there $\hat{a}, \hat{b}$ close to a, b s.t.
my output $\tilde{y}$ is exactly $\hat{a}\hat{b}^T$?

$\tilde{y}$ is a matrix of rank 1 ($\Rightarrow$ it can be written as the outer product of two vectors)
 $A = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$
 $r = \text{rank}(A)$

NO, we can't $\Rightarrow$ not backward stable.

Def: We say that an algorithm is BACKWARD STABLE (on the input x) if the result $\tilde{y}(x)$ it computes in machine arithmetic can be seen as the exact solution of a slightly perturbed input $y(\hat{x})$, where

$$\frac{\|\hat{x} - x\|}{\|x\|} = O(u)$$

if an algorithm is backward stable, then

$$\frac{\|\tilde{y} - y\|}{\|y\|} = \frac{\|y(\hat{x}) - y(x)\|}{\|y(x)\|} \leq \underset{\substack{\uparrow \\ \text{backward} \\ \text{stability}}}{\kappa_{\text{rel}}(f, x)} \cdot \frac{\|\hat{x} - x\|}{\|x\|} \leq \kappa_{\text{rel}}(f, x) \cdot O(u)$$

$\rightarrow = O(u)$ by backward stability

BACKWARD STABLE $\Rightarrow$ STABLE

"A posteriori" stability check

Linear systems:

$$Ax = b \quad A \in \mathbb{R}^{n \times n} \quad b \in \mathbb{R}^n$$

$\rightsquigarrow$ computed solution $\tilde{x}$

$$\text{Let } \tilde{r} = A\tilde{x} - b \Rightarrow A\tilde{x} = \underbrace{b + \tilde{r}}_{\hat{b}}$$

$\Rightarrow$ the residual $\tilde{r}$ solves the perturbed linear system

$$A\tilde{x} = \hat{b}, \text{ with } \hat{b} = b + \tilde{r}.$$

$$\Rightarrow \frac{\|\tilde{x} - x\|}{\|x\|} \leq \kappa(A) \cdot \frac{\|\hat{b} - b\|}{\|b\|} = \kappa(A) \cdot \frac{\|\tilde{r}\|}{\|b\|}$$

If the returned sol. $\tilde{x}$ is such that $\kappa(A) \cdot \frac{\|\tilde{r}\|}{\|b\|}$ is small then we automatically know that $\tilde{x}$ is a good approx. of x .

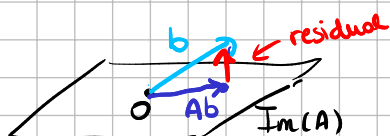
(let's ignore the fact that $\tilde{r}$ and other quantities like $\kappa(A)$ and $\|b\|$ are computed in machine arithmetic, since the error arising from these operations is usually negligible with respect to the rest).

Least-squares problems:

$$\min_{x \in \mathbb{R}^m} \|Ax - b\| \quad A \in \mathbb{R}^{n \times m} \quad b \in \mathbb{R}^n$$

$\leadsto$ computed solution $\tilde{x}$

$$\leadsto \text{residual } \tilde{r} := A\tilde{x} - b$$



$\hookrightarrow$ problem: even for exact solution, this is not small, in general...

$$\text{what we do have: } f(x) = \frac{1}{2} \|Ax - b\|^2$$

$$\nabla f(\tilde{x}) = A^T A \tilde{x} - A^T b = A^T (A\tilde{x} - b) = A^T \tilde{r}$$

$\leadsto$ we expect $\nabla f(\tilde{x}) = A^T \tilde{r}$ to be small.

x is the solution to $A^T A x = A^T b$ (normal equations)

$\tilde{x}$ is approximate sol. to this linear system

and the residual is $A^T A \tilde{x} - A^T b = A^T \tilde{r}$ $\rightarrow$ residual for the LS problem

$$\Rightarrow \frac{\|\tilde{x} - x\|}{\|x\|} \leq \underbrace{\kappa(A^T A)}_{\text{residual for the normal equations}} \cdot \frac{\|A^T \tilde{r}\|}{\|A^T b\|}$$

$\hookrightarrow$ Ex: show that $\kappa(A^T A) = (\kappa(A))^2$.

$\hookrightarrow$ could be much worse than $\kappa(A)$.

Trick: x solves LS problem $\Leftrightarrow$

$$(*) \quad \begin{bmatrix} \alpha I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} -\frac{1}{\alpha} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix} \quad \forall \alpha \neq 0$$

$$\begin{cases} \alpha I \cdot \left(-\frac{1}{\alpha} r\right) + Ax = b \\ -\frac{1}{\alpha} A^T r = 0 \end{cases} \quad \begin{cases} r = Ax - b \\ A^T r = 0 \end{cases}$$

Lemma: For an approximate solution $\tilde{x}$ of the LS problem, $\tilde{r} = A\tilde{x} - b$, then for all $\alpha > 0$ we have

$$\leadsto \begin{cases} A^T (Ax - b) = 0 \\ A^T A x = A^T b \end{cases}$$

$$\frac{\|\tilde{x} - x\|}{\|[-\frac{1}{\alpha} r, x]\|} \leq \kappa \left(\begin{bmatrix} \alpha I & A \\ A^T & 0 \end{bmatrix} \right) \cdot \frac{\|\frac{1}{\alpha} A^T \tilde{r}\|}{\|b\|}$$

why? Apply analysis for linear systems to (*)

$$\text{Ex: } \kappa \left(\begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \right) = ?$$